BELLOMASSI

INFRASTRUCTURE AND DEVOPS

Hosting Options — Cloud vs On-Premises Comparison

Bellomassi Order-to-Delivery Management Platform

Document ID BM-INF-002

Document control

Field	Value
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> **Status labels used in this document:** VERIFIED (confirmed in the current source code or existing documents) · PROTOTYPE (works in the React/LocalStorage prototype with mock or simulated behaviour) · PROPOSED (recommended production design, not built) · PENDING CONFIRMATION (needs a client decision) · NOT IMPLEMENTED · NOT VERIFIED.

Purpose

Compare AWS, Azure, DigitalOcean and on-premises hosting to support the client's hosting decision.

Scope

Qualitative comparison; costs are indicative estimates in BM-INF-004, not quotations. Decision PENDING CONFIRMATION.

Definitions

Other terms: see **BM-GLO-001** (Project Glossary).

References

- **BM-INF-003** (Infrastructure, Network Architecture and Server Specification)
- **BM-INF-004** (Infrastructure Cost Estimation (ESTIMATE))

1. Comparison

Criterion	AWS	Azure	DigitalOcean	On-premises
Middle East region	Yes (UAE, Bahrain)	Yes (UAE North / Central)	No ME region (nearest Bangalore / Frankfurt)	Client data centre
India region	Yes (Mumbai, Hyderabad)	Yes (Pune, Chennai…)	Bangalore	Client data centre
Managed PostgreSQL	RDS / Aurora	Azure Database for PostgreSQL	Managed Databases	Self-managed
Managed Redis	ElastiCache	Azure Cache for Redis	Managed Redis	Self-managed
Containers	ECS / Fargate, EKS	Container Apps, AKS	App Platform, DOKS	Docker / Kubernetes on VMs
Object storage	S3	Blob Storage	Spaces	MinIO
Security / compliance tooling	Extensive	Extensive; strong with Microsoft 365 / Entra ID	Basic	Client responsibility
Operations effort	Medium	Medium	Low	High
Cost level (indicative)	Medium	Medium	Low	High upfront, low monthly
Data residency	In-country regions available	In-country regions available	Limited	Full control
Skills needed	AWS	Azure	Generalist	Infrastructure team

2. Recommendation — PROPOSED, PENDING CONFIRMATION

1. If the client uses Microsoft 365 / Entra ID → **Azure UAE North** (SSO, same tenant).
2. Otherwise → **AWS me-central-1 (UAE)** with managed services.
3. DigitalOcean only for non-production (test / demo) because of region and compliance limits.
4. On-premises only if data-residency policy requires it and an operations team exists.

If the operating country is India, use the equivalent Indian regions.

Revision history

Version	Date	Author	Description
1.0	2026-11-12	Documentation team	Initial draft prepared from analysis of the bellomassi-premium-ui source code and existing project documents.